MTS Tamilnadu · 2020

31 Ans A = 24 days → 1 U/day
B = 6 days → 4 U/d
C = 12 days → 2 U/d
24 Units. Let L.C.M = Total work

(A+B+C)'s one day work = 1+4+2 = 7 Units
Time taken by them to complete 24 units of work
$= \frac{24}{7} = 3\frac{3}{7}$ days.

32 Ans ① A+B = 8 days → 3 U/day
② B+C = 12 days → 2 U/d
③ A+B+C = 6 days → 4 U/day
24 Units. Let L.C.M = Total work

Subtract ① from ③ we get

$$A+B+C = 4$$
$$-\ A+B = 3$$
$$C = 1 \text{ Unit/day} \cdots ④$$

and subtract ② from ③ we get

$$A+B+C = 4$$
$$-\ B+C = 2$$
$$A = 2 \text{ U/day} \cdots ⑤$$

From ④ and ⑤ (A+C)'s one day work = 2+1 = 3 Unit
Time taken by them to complete 24 units of work
$= \frac{24}{3} = 8$ days.

33 Ans Speed = 5 k.m/h $= 5 \times \frac{5}{18} = \frac{25}{18}$ m/s.

Time = 15 minutes = 15 × 60 = 900 second

distance = Speed × Time

$= \frac{25}{18} \times 900 = 25 \times 50 = 1250$ m.

POSTAL A B C

41 Ans $112 \times 5^4 = 112 \times 5 \times 5 \times 5 \times 5$

$= 625 \times 112 = 70000$ Ans

42 Ans $217 \times 217 + 183 \times 183$

$= 47,089 + 33489$

$= 80578.$

43 Ans Average Of 5 numbers = 27

Total Value = $27 \times 5 = 135$

When one number excluded

Average of remaining 4 numbers = 25

Total Value of 4 numbers = $25 \times 4 = 100$

So, excluded number = $135 - 100 = 35$ Ans

44 Ans 88% Of 370 (+) 24% of $210 - x = 118.$

$\Rightarrow \frac{88}{100} \times 370 + \frac{24}{100} \times 210 - x = 118.$

$\Rightarrow 325.6 + 50.4 - 118 = x.$

$\Rightarrow x = 376 - 118 = 258$ Ans

45 Ans Total Cost price = $4700 + 800 = 5500$ R.S

Selling price = 5800

Profit = $5800 - 5500 = 300$ R.S.

Profit % = $\frac{\text{Profit}}{\text{C.P}} \times 100 = \frac{300}{5500} \times 100 = \frac{300}{55} = \frac{60}{11}$

$= 5\frac{5}{11}\%.$

POSTAL A B C

46 Ans Distance = 10 k.m

Time = 12 minut = $\frac{12}{60} = \frac{1}{5}$ hour.

Speed = $\frac{\text{Distance}}{\text{Time}} = \frac{10}{\frac{1}{5}} = 50$ k.m/hour.

Now when speed reduced by 5 k.m/hour.

Then new speed = 50-5 = 45 k.m/hour.

Time = $\frac{\text{Distance}}{\text{Speed}} = \frac{10}{45}$ hours = $\frac{10}{45} \times 60$ minutes

= $\frac{10 \times 4}{3} = \frac{40}{3} = 13$ minut $\frac{1}{3}$ minut

= 13 minut $\frac{1}{3} \times 60$ second

= 13 minutes 20 seconds.

47 Ans Principal = 16800 R.S Rate = $6\frac{1}{4} = \frac{25}{4}$ %

Time = 9 months = $\frac{9}{12}$ years = $\frac{3}{4}$ years.

SI = $\frac{P \times R \times T}{100} = \frac{16800 \times 25 \times 3}{100 \times 4 \times 4}$

= 787.50 Ans

POSTAL A B C